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(57) Abstract:

The agentic system described here acts as a digital assistant that helps users remember their past activities across different digital platforms. It works by gathering information from sources like email, documents, browsers, and app logs. The system then analyses this data, looking for patterns in timing and meaning, so that it can connect related pieces of information even if they are scattered across different places. By doing this, it creates a network of related events that reflect how users actually experienced them, making it much easier to recall past communications and actions. The correlation engine uses temporal proximity analysis, semantic similarity modelling, and behavioural pattern inference to build a structured event graph of inferred user experiences. The reconstruction engine then applies this graph to generate a coherent, timeline-based narrative of past digital activity. This narrative is presented through an interactive interface that allows user inspection, refinement, and adaptive quer

Agentic System for Contextual Digital Memory Reconstruction Using Multi-Source Activity Correlation

Abstract:

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The correlation engine uses temporal proximity analysis, semantic similarity modelling, and behavioural pattern inference to build a structured event graph of inferred user experiences. The reconstruction engine then applies this graph to generate a coherent, timeline-based narrative of past digital activity. This narrative is presented through an interactive interface that allows user inspection, refinement, and adaptive query expansion.

The system operates as a persistent background service within a user's personal computer, maintaining activity data locally unless configured otherwise. It functions as an externalised digital episodic memory layer. By integrating agent orchestration, multi-source activity correlation, and narrative synthesis, the system enables situational recall through high-level contextual cues rather than explicit identifiers. This approach improves productivity, reduces redundant effort, and enhances the efficiency of the human–computer interaction.

Introduction:

The agentic-event-graph mechanism is part of an agentic computing system that helps users remember and understand their digital activities. As people interact daily with platforms such as emails, documents, websites, and social media, important information often becomes scattered and difficult to recall.

The system uses autonomous software agents that run in the background to collect and analyse fragments of user activities. By examining contextual similarity, timing, and user behaviour, these agents identify connections between actions. Recognised as part of the same event. These linked activities are organised into an event graph that visually depicts relationships between user actions.

A reconstruction engine then converts this graph into a clear, easy-to-understand narrative presented through an interactive recall interface. Acting as a persistent digital memory layer, the system helps users quickly revisit past experiences and retrieve important information across academic, professional, and personal tasks.

Description:

Background of the Invention:

The current information retrieval infrastructure is mainly document-centric and relies on explicit identifiers like keywords, filenames, participants, or timestamps. In reality, people tend to remember only high-level situational information about an experience, such as a topic of a conversation, a rough time, or the simultaneous use of multiple apps, without remembering the actual query keywords. The lack of support for reconstructing experiences from disjoint activity traces is a problem with traditional keyword search, fixed filters, or single-service history views.

Summary of the Invention:

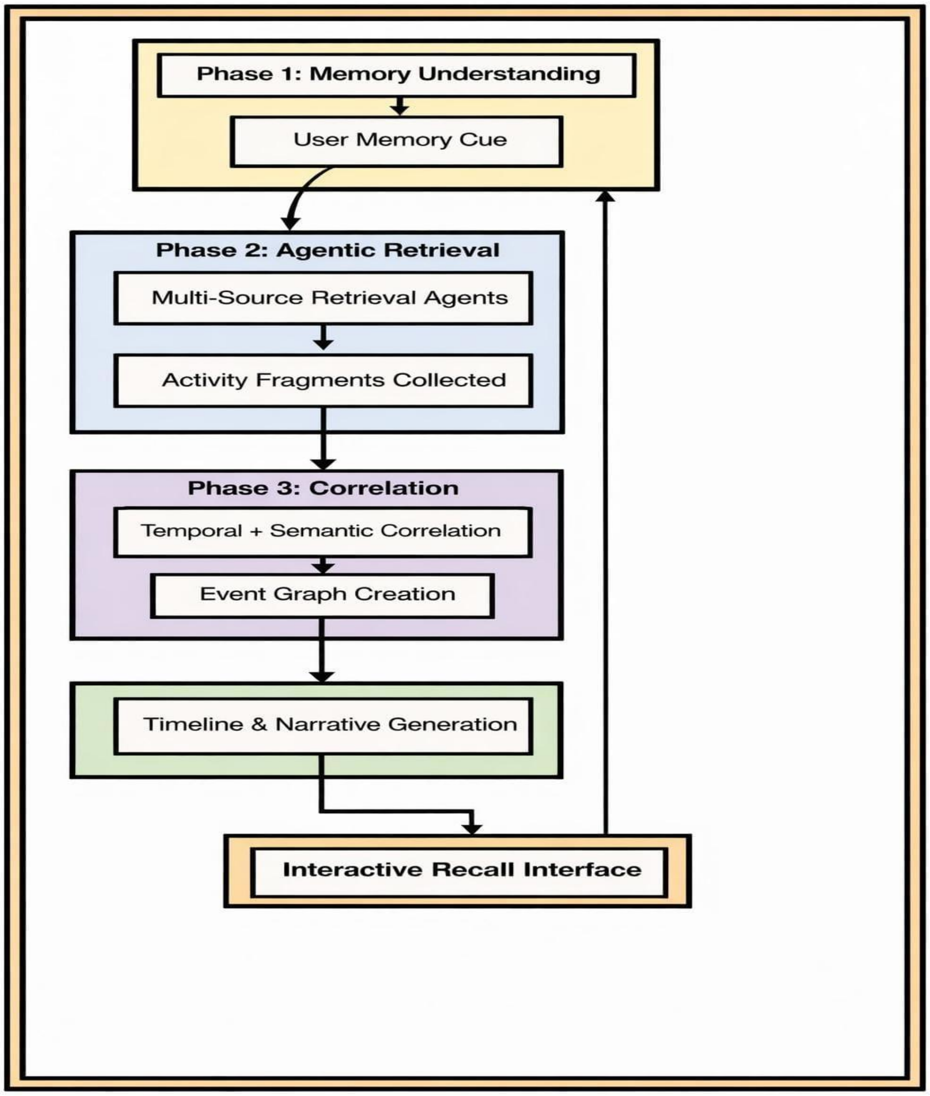
The invention discloses a system and method for contextual digital memory reconstruction using an agentic architecture that coordinates multiple autonomous software agents to understand and rebuild previous user experiences from digital activity data across a wide range of sources. Instead of returning isolated documents or messages, the system generates an experiential event representation that links actions, applications, and information objects into a logical narrative corresponding to a user's recollection.

The system receives a user-provided memory cue, usually as natural-language representing a vague or partial recollection of a past digital experience. A memory interpretation component processes the cue to extract contextual signals such as topics, entities, approximate time indicators, mentioned applications, and behavioural hints. It generates one or more structured memory hypotheses for further processing or documentation.

In another aspect, an agentic retrieval layer comprises a plurality of specialised agents configured to operate over different activity sources, such as operating system logs, browser history, email messages, messaging conversations, calendars, documents, and media files. Based on the memory hypotheses, the agents retrieve candidate activity fragments from their respective sources and supply them to a correlation engine. The correlation engine applies time-based correlation, semantic similarity analysis, and behavioural pattern detection to link related activity fragments into a unified event graph representing inferred experiential events.

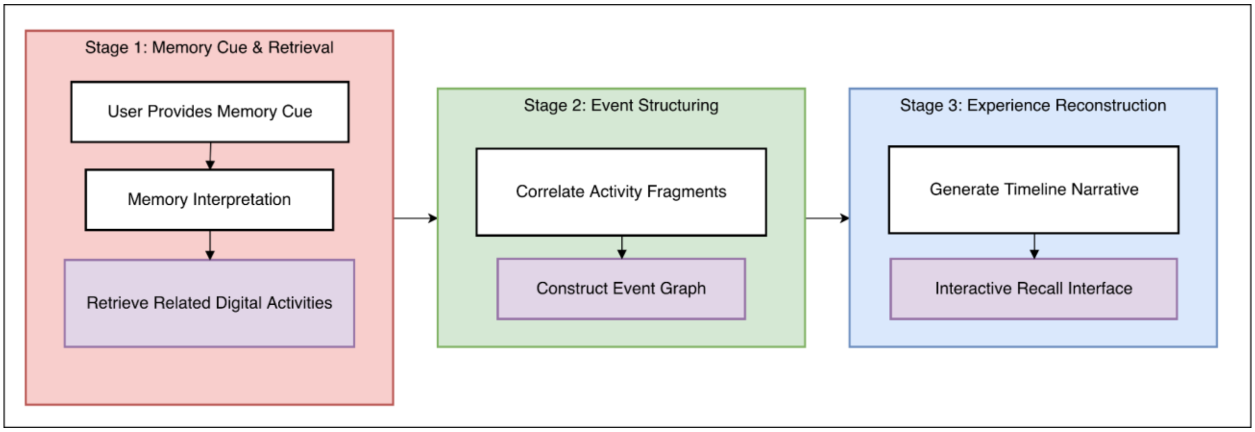
A reconstruction engine uses the event graph and produces an interactive and user-friendly reconstruction of the digital experience, including a temporal sequence of key actions, associated content items, involved participants, and relevant contextual data. The narrative is presented to the user via an interactive recall interface that allows inspection, refinement, and follow-up queries. The system operates as a persistent background service in the user's personal computer, continuously ingesting activity signals and incrementally updating the event graph to maintain an externalised digital episodic memory layer.

Figure 1: Overall Architectural Flow



The figure-1 shows the complete multi-layered workflow of the system. The process begins with a natural-language memory cue, which is interpreted into structured hypotheses. Retrieval agents collect relevant activity fragments from various sources. These fragments are correlated and organised into an event graph. The system then reconstructs a chronological narrative, presented through an interactive recall interface that supports refinement and real-time updates.

Figure 2: Simplified overview of the contextual digital memory reconstruction process.



The figure-2 provides a simplified three-stage overview of the system. The first stage interprets the user's memory cue and retrieves related digital activities. The second stage structures the collected data into a graph of events. The final stage reconstructs the experience into a narrative form and presents it through an interactive interface.

Use Cases:

1. The system helps knowledge workers, students, and professionals recover complex project backgrounds, reducing repeated work and improving productivity when switching between multiple digital tools.
2. The system supports individuals with mild cognitive impairment or attention difficulties by acting as an externalised episodic memory, enabling them to recall what they were doing, with whom, and which information resources they used.
3. The system can assist in digital investigation and audit scenarios, where reconstructing a person's sequence of actions across applications is necessary to understand decision-making or trace the origin of errors.
4. The system improves general human-computer interaction by allowing users to recall past activities using natural, situational language instead of technical search parameters, making digital environments more accessible to non-expert users.

Claim:

1. A computer-implemented system for contextual digital memory reconstruction executed by one or more processors and comprising: an activity ingestion module configured to collect user activity data from a plurality of heterogeneous digital sources; a memory interpretation module configured to receive a natural-language memory cue and generate one or more structured memory hypotheses; an agent orchestration module configured to coordinate a plurality of retrieval agents to obtain activity fragments from the heterogeneous digital sources based on the one or more structured memory hypotheses; a correlation module configured to construct an event graph by linking the activity fragments using temporal proximity, semantic similarity, and behavioural relationships; and a reconstruction engine configured to generate, from at least a portion of the event graph, a narrative representation of a reconstructed digital experience for presentation via an interactive recall interface, wherein the event graph enables cross-application contextual reconstruction beyond keyword-based retrieval systems.
2. The system of claim 1, wherein the heterogeneous digital sources comprise at least one of operating system event logs, web browser history, email messages, messaging conversations, calendar events, productivity documents, or media interaction records.
3. The system of claim 1, wherein the agent orchestration module is further configured to progressively refine the one or more structured memory hypotheses based on certainty scores, relevance metrics, or context-based annotations returned by the plurality of retrieval agents, and to initiate additional retrieval operations using the refined hypotheses.
4. The system of claim 1, wherein the correlation module is further configured to incrementally update the event graph in response to newly ingested activity data, such that the event graph functions as a continuously maintained, externally accessible digital episodic memory for at least one user.
5. The system of claim 1, wherein the reconstruction engine is further configured to dynamically modify the narrative representation in real time in response to follow-up queries received through the interactive recall interface by re-weighting, filtering, or re-ordering nodes and edges of the event graph to generate an updated reconstructed digital experience.

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